

```
>> dynare mcmc.mod
Starting Dynare (version 6.4).
Calling Dynare with arguments: none
Starting preprocessing of the model file ...
Found 23 equation(s).
Evaluating expressions...
Computing static model derivatives (order 2).
Computing static model derivatives w.r.t. parameters (order 2).
Normalizing the static model...
Finding the optimal block decomposition of the static model...
5 block(s) found:
  3 recursive block(s) and 2 simultaneous block(s).
  the largest simultaneous block has 16 equation(s)
    and 16 feedback variable(s).
Computing dynamic model derivatives (order 2).
Computing dynamic model derivatives w.r.t. parameters (order 2).
Normalizing the dynamic model...
Finding the optimal block decomposition of the dynamic model...
5 block(s) found:
  3 recursive block(s) and 2 simultaneous block(s).
  the largest simultaneous block has 16 equation(s)
    and 16 feedback variable(s).

Preprocessing completed.
Preprocessing time: 0h00m00s.
```

STEADY-STATE RESULTS:

q	32.8507
k	4.41224
R	1.01351
b	143.013
x	0.0406367
varphi	0.0303425
mu	0.00686924
kp	2.20776
chi	1.02177
nu	0.00432944
eta	1.28813
zeta	1.02177
N	41.8193
phi	3.41978
Ne	41.5333
Nn	0.286025
xp	6.22046
bp	-101.193
Y	3.45318
eb	0
N_obs	0
Y_obs	0
q_obs	0

EIGENVALUES:

Modulus	Real	Imaginary
0	0	0
3.882e-18	-3.882e-18	0
1.364e-17	1.364e-17	0
1.027e-14	-1.027e-14	0
4.029e-09	5.157e-15	4.029e-09

4.029e-09	5.157e-15	-4.029e-09
0.1469	0.1469	0
0.9804	0.9804	0
1.018	1.018	0
1.02	1.02	0
1.022	1.022	0
1.038	1.038	0
3.365	-3.365	0
1.193e+20	-1.193e+20	0
2.084e+20	-2.084e+20	0

There are 7 eigenvalue(s) larger than 1 in modulus for 7 forward-looking variable(s)
The rank condition is verified.

===== Identification Analysis =====

Testing prior mean

Note that differences in the criteria could be due to numerical settings,
numerical errors or the method used to find problematic parameter sets.

Settings:

Derivation mode for Jacobians:	Analytic using sylvester equations
Method to find problematic parameters:	Nullspace and multicorrelation
coefficients	
Normalize Jacobians:	Yes
Tolerance level for rank computations:	robust
Tolerance level for selecting nonzero columns:	1e-08
Tolerance level for selecting nonzero singular values:	1e-03

REDUCED-FORM:

All parameters are identified in the Jacobian of steady state and reduced-form solution matrices (rank(Tau) is full with tol = robust).

MINIMAL SYSTEM (KOMUNJER AND NG, 2011):

All parameters are identified in the Jacobian of steady state and minimal system (rank(Deltabar) is full with tol = robust).

SPECTRUM (QU AND TKACHENKO, 2012):

All parameters are identified in the Jacobian of mean and spectrum (rank(Gbar) is full with tol = robust).

MOMENTS (ISKREV, 2010):

All parameters are identified in the Jacobian of first two moments (rank(J) is full with tol = robust).

==== Identification analysis completed ====

You did not declare endogenous variables after the estimation/calib_smoother command.
Initial value of the log posterior (or likelihood): 283.1433

```

-----
f at the beginning of new iteration,      -283.1432772161
Predicted improvement:      45.651883948
lambda =      1; f =      -283.1394258
lambda =      0.33333; f =      -283.1430140
lambda =      0.11111; f =      -283.1432763

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lambda = 0.037037; f = -282.4742566
lambda = 0.012346; f = -283.9719129
lambda = 0.023866; f = -283.9916769
Norm of dx 0.095553
----
Improvement on iteration 1 = 0.848399685
-----
f at the beginning of new iteration, -283.9916769014
Predicted improvement: 28.816858168
lambda = 1; f = -283.9900215
lambda = 0.33333; f = -270.7251790
lambda = 0.11111; f = -285.0017144
lambda = 0.037037; f = -285.4199562
Norm of dx 0.096149
----
Improvement on iteration 2 = 1.428279316
-----
f at the beginning of new iteration, -285.4199562179
Predicted improvement: 3.781889335
lambda = 1; f = -285.4198129
lambda = 0.33333; f = -255.5192748
lambda = 0.11111; f = -284.4780324
lambda = 0.037037; f = -285.5331493
Norm of dx 0.029897
----
Improvement on iteration 3 = 0.113193085
-----
f at the beginning of new iteration, -285.5331493025
Predicted improvement: 0.270322477
lambda = 1; f = -285.8352645
Norm of dx 0.0021123
----
Improvement on iteration 4 = 0.302115156
-----
f at the beginning of new iteration, -285.8352644581
Predicted improvement: 0.012240011
lambda = 1; f = -285.8515550
Norm of dx 0.0010983
----
Improvement on iteration 5 = 0.016290509
-----
f at the beginning of new iteration, -285.8515549672
Predicted improvement: 0.006460934
lambda = 1; f = -285.8625119
lambda = 1.9332; f = -285.8692033
lambda = 3.7372; f = -285.8726873
Norm of dx 0.0016112
----
Improvement on iteration 6 = 0.021132344
-----
f at the beginning of new iteration, -285.8726873112
Predicted improvement: 0.014337834
lambda = 1; f = -285.9010780
lambda = 1.9332; f = -285.9270679
lambda = 3.7372; f = -285.9759179
lambda = 7.2247; f = -286.0650561
lambda = 13.967; f = -286.2169012
lambda = 27; f = -286.4287408

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lambda =      52.196; f =      -286.4908885
Norm of dx  0.0028978
----
Improvement on iteration 7 =      0.618201162
-----
f at the beginning of new iteration,      -286.4908884727
Predicted improvement:      0.723612758
lambda =      1; f =      -287.8806607
lambda =      1.9332; f =      -289.0714210
lambda =      3.7372; f =      -291.0663945
lambda =      7.2247; f =      -293.6139052
Norm of dx  0.13892
----
Improvement on iteration 8 =      7.123016750
-----
f at the beginning of new iteration,      -293.6139052229
Predicted improvement:      3.992292142
lambda =      1; f =      -292.4839947
lambda =      0.33333; f =      -295.4912754
lambda =      0.64439; f =      -294.3124275
lambda =      0.4339; f =      -295.5802865
Norm of dx  1.4337
----
Improvement on iteration 9 =      1.966381300
-----
f at the beginning of new iteration,      -295.5802865230
Predicted improvement:      0.567640725
lambda =      1; f =      -296.2848664
Norm of dx  0.039264
----
Improvement on iteration 10 =      0.704579858
-----
f at the beginning of new iteration,      -296.2848663812
Predicted improvement:      0.032476452
lambda =      1; f =      -296.3093253
Norm of dx  0.0047976
----
Improvement on iteration 11 =      0.024458872
-----
f at the beginning of new iteration,      -296.3093252534
Predicted improvement:      0.002951950
lambda =      1; f =      -296.3116083
Norm of dx  0.01472
----
Improvement on iteration 12 =      0.002283053
-----
f at the beginning of new iteration,      -296.3116083065
Predicted improvement:      0.000572913
lambda =      1; f =      -296.3122613
Norm of dx  0.0069836
----
Improvement on iteration 13 =      0.000652944
-----
f at the beginning of new iteration,      -296.3122612503
Predicted improvement:      0.000328045
lambda =      1; f =      -296.3128768
lambda =      1.9332; f =      -296.3133801
lambda =      3.7372; f =      -296.3141576

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lambda =      7.2247; f =      -296.3149227
Norm of dx  0.0017595
----
Improvement on iteration 14 =      0.002661469
-----
f at the beginning of new iteration,      -296.3149227193
Predicted improvement:      0.002632863
lambda =      1; f =      -296.3201323
lambda =      1.9332; f =      -296.3248947
lambda =      3.7372; f =      -296.3338287
lambda =      7.2247; f =      -296.3500712
lambda =      13.967; f =      -296.3775588
lambda =      27; f =      -296.4155521
lambda =      52.196; f =      -296.4281917
Norm of dx  0.0041469
----
Improvement on iteration 15 =      0.113269010
-----
f at the beginning of new iteration,      -296.4281917291
Predicted improvement:      0.118640026
lambda =      1; f =      -296.6398435
lambda =      1.9332; f =      -296.7929848
lambda =      3.7372; f =      -296.9776031
Norm of dx  0.087785
----
Improvement on iteration 16 =      0.549411326
-----
f at the beginning of new iteration,      -296.9776030546
Predicted improvement:      0.157907608
lambda =      1; f =      -297.1775482
Norm of dx  0.36834
----
Improvement on iteration 17 =      0.199945127
-----
f at the beginning of new iteration,      -297.1775481813
Predicted improvement:      0.022284333
lambda =      1; f =      -297.2025029
Norm of dx  0.15217
----
Improvement on iteration 18 =      0.024954682
-----
f at the beginning of new iteration,      -297.2025028637
Correct for low angle: 0.00474477
Predicted improvement:      0.000680923
lambda =      1; f =      -297.2031895
Norm of dx  0.017635
----
Improvement on iteration 19 =      0.000686613
-----
f at the beginning of new iteration,      -297.2031894764
Predicted improvement:      0.000005808
lambda =      1; f =      -297.2031972
Norm of dx  0.00076251
----
Improvement on iteration 20 =      0.000007736
-----
f at the beginning of new iteration,      -297.2031972122
Predicted improvement:      0.000000630

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lambda =          1; f =          -297.2031982
lambda =       1.9332; f =          -297.2031980
Norm of dx 0.00032025
----
Improvement on iteration 21 =          0.000000992
-----
f at the beginning of new iteration,          -297.2031982046
Correct for low angle: 0.00300597
Predicted improvement:          0.000000002
lambda =          1; f =          -297.2031982
lambda =       1.9332; f =          -297.2031983
lambda =       3.7372; f =          -297.2031983
lambda =       7.2247; f =          -297.2031983
lambda =      13.967; f =          -297.2031982
lambda =        27; f =          -297.2031974
lambda =      18.18; f =          -297.2031981
lambda =      12.242; f =          -297.2031983
lambda =      15.52; f =          -297.2031982
lambda =      13.46; f =          -297.2031983
lambda =      14.661; f =          -297.2031982
lambda =      13.928; f =          -297.2031982
lambda =      14.363; f =          -297.2031982
Norm of dx 1.8707e-05
----
Improvement on iteration 22 =          0.000000142
warning: possible inaccuracy in H matrix
-----
f at the beginning of new iteration,          -297.2031983466
Correct for low angle: 0.00467279
Predicted improvement:          0.000000121
lambda =          1; f =          -297.2031982
lambda =       0.33333; f =          -297.2031983
lambda =       0.11111; f =          -297.2031983
lambda =       0.037037; f =          -297.2031983
lambda =       0.012346; f =          -297.2031983
lambda =       0.0041152; f =          -297.2031983
lambda =       0.0013717; f =          -297.2031983
lambda = 0.00045725; f =          -297.2031983
lambda = 0.00015242; f =          -297.2031983
lambda = 5.0805e-05; f =          -297.2031983
lambda = 1.6935e-05; f =          -297.2031983
lambda = 5.645e-06; f =          -297.2031983
lambda = 1.8817e-06; f =          -297.2031983
lambda = 6.2723e-07; f =          -297.2031983
lambda = 1.2125e-06; f =          -297.2031983
lambda = 8.1646e-07; f =          -297.2031983
lambda = 6.4398e-07; f =          -297.2031983
lambda = 6.3221e-07; f =          -297.2031983
Norm of dx 0.0001137
Cliff. Perturbing search direction.
Predicted improvement:          0.000000228
lambda =          1; f =          -297.2031979
lambda =       0.33333; f =          -297.2031983
lambda =       0.11111; f =          -297.2031983
lambda =       0.037037; f =          -297.2031983
lambda =       0.012346; f =          -297.2031983
lambda =       0.0041152; f =          -297.2031983
lambda =       0.0013717; f =          -297.2031983

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```

lambda = 0.00045725; f = -297.2031983
lambda = 0.00015242; f = -297.2031983
lambda = 5.0805e-05; f = -297.2031983
lambda = 1.6935e-05; f = -297.2031983
lambda = 5.645e-06; f = -297.2031983
lambda = 1.8817e-06; f = -297.2031983
lambda = 6.2723e-07; f = -297.2031983
lambda = 2.0908e-07; f = -297.2031983
lambda = 6.9692e-08; f = -297.2031983
lambda = 2.3231e-08; f = -297.2031983
lambda = 7.7435e-09; f = -297.2031983
lambda = 2.5812e-09; f = -297.2031983
lambda = 4.9899e-09; f = -297.2031983
lambda = 3.3599e-09; f = -297.2031983
lambda = 4.2598e-09; f = -297.2031983
lambda = 3.6944e-09; f = -297.2031983
lambda = 3.2042e-09; f = -297.2031983
lambda = 2.7789e-09; f = -297.2031983
lambda = 3.0268e-09; f = -297.2031983
lambda = 3.2967e-09; f = -297.2031983
lambda = 3.5907e-09; f = -297.2031983
lambda = 3.911e-09; f = -297.2031983
lambda = 3.7156e-09; f = -297.2031983
lambda = 3.5299e-09; f = -297.2031983
lambda = 3.6401e-09; f = -297.2031983
lambda = 3.5736e-09; f = -297.2031983
lambda = 3.5083e-09; f = -297.2031983
lambda = 3.4441e-09; f = -297.2031983
lambda = 3.3811e-09; f = -297.2031983
lambda = 3.3193e-09; f = -297.2031983
lambda = 3.2586e-09; f = -297.2031983
lambda = 3.1991e-09; f = -297.2031983
lambda = 3.2347e-09; f = -297.2031983
Norm of dx 7.1971e-05
Cliff again. Try traversing
Predicted improvement: 0.000000000
lambda = 1; f = -297.2031983
lambda = 0.33333; f = -297.2031983
lambda = 0.11111; f = -297.2031983
lambda = 0.037037; f = -297.2031983
lambda = 0.071599; f = -297.2031983
lambda = 0.048211; f = -297.2031983
Norm of dx 7.8705e-06
----
Improvement on iteration 23 = 0.000000000
improvement < crit termination
back and forth on step length never finished

Final value of minus the log posterior (or likelihood):-297.203198

MODE CHECK

Fval obtained by the optimization routine: -297.203198

RESULTS FROM POSTERIOR ESTIMATION
parameters
prior mean mode s.d. prior pstdev

```

gamma_q	1.5721	0.9002	0.7602	norm 1.0000
gamma_bY	1.9539	2.1293	0.3826	norm 1.0000

standard deviation of shocks

	prior mean	mode	s.d.	prior pstdev
eN	0.0545	0.0536	0.0037	invg 0.0050
eY	0.0172	0.0171	0.0016	invg 0.0050
eq	0.0084	0.0082	0.0011	invg 0.0050

Log data density [Laplace approximation] is 281.155226.

Estimation::mcmc: Multiple chains mode.

Estimation::mcmc: Old mh-files successfully erased!

Estimation::mcmc: Old metropolis.log file successfully erased!

Estimation::mcmc: Creation of a new metropolis.log file.

Estimation::mcmc: Searching for initial values...

Estimation::mcmc: Initial values found!

Estimation::mcmc: Write details about the MCMC... Ok!

Estimation::mcmc: Details about the MCMC are available in mcmc/metropolis\mcmc_mh_history_0.mat

Estimation::mcmc: Number of mh files: 1 per block.

Estimation::mcmc: Total number of generated files: 2.

Estimation::mcmc: Total number of iterations: 125000.

Estimation::mcmc: Current acceptance ratio per chain:

Chain 1: 47.6728%

Chain 2: 47.4984%

Estimation::mcmc: Total number of MH draws per chain: 125000.

Estimation::mcmc: Total number of generated MH files: 1.

Estimation::mcmc: I'll use mh-files 1 to 1.

Estimation::mcmc: In MH-file number 1 I'll start at line 25001.

Estimation::mcmc: Finally I keep 100000 draws per chain.

MCMC Inefficiency factors per block

Parameter	Block 1	Block 2
SE_eN	27.966	23.647
SE_eY	25.996	27.389
SE_eq	35.333	29.466
gamma_q	22.907	19.682
gamma_bY	25.423	23.650

Convergence diagnostics results for chain 1.

Geweke (1992) Convergence Tests, based on means of draws 25000 to 45000 vs 75000 to 125000 for chain 1.

p-values are for Chi2-test for equality of means.

Parameter	Post. Mean	Post. Std	p-val No Taper	p-val 4% Taper	p-val 8% Taper
p-val 15% Taper					
SE_eN	0.055	0.004	0.059	0.698	
0.687	0.675				
SE_eY	0.017	0.002	0.000	0.060	

0.031	0.031				
SE_eq	0.009	0.001	0.000	0.123	
0.126	0.101				
gamma_q	0.854	0.760	0.351	0.829	
0.804	0.773				
gamma_bY	2.234	0.393	0.000	0.365	
0.308	0.254				

Convergence diagnostics results for chain 2.

Geweke (1992) Convergence Tests, based on means of draws 25000 to 45000 vs 75000 to 125000 for chain 2.

p-values are for Chi2-test for equality of means.

Parameter	Post. Mean	Post. Std	p-val No Taper	p-val 4% Taper	p-val 8% Taper
p-val 15% Taper					
SE_eN	0.055	0.004	0.000	0.482	
0.479	0.499				
SE_eY	0.017	0.002	0.612	0.918	
0.913	0.917				
SE_eq	0.009	0.001	0.000	0.366	
0.381	0.410				
gamma_q	0.851	0.752	0.000	0.092	
0.076	0.078				
gamma_bY	2.238	0.389	0.000	0.046	
0.041	0.017				

Univariate convergence diagnostic, Brooks and Gelman (1998):

Parameter 1... Done!
 Parameter 2... Done!
 Parameter 3... Done!
 Parameter 4... Done!
 Parameter 5... Done!

marginal density: I'm computing the posterior mean and covariance... Done!

marginal density: I'm computing the posterior log marginal density (modified harmonic mean)... Done!

ESTIMATION RESULTS

Log data density (Modified Harmonic Mean) is 281.238371.

parameters

	prior mean	post. mean	90% HPD interval		prior	pstdev
gamma_q	1.572	0.8540	-0.3787	2.0993	norm	1.0000
gamma_bY	1.954	2.2375	1.5913	2.8747	norm	1.0000

standard deviation of shocks

	prior mean	post. mean	90% HPD interval		prior	pstdev
eN	0.054	0.0548	0.0485	0.0610	inv	0.0050
eY	0.017	0.0175	0.0147	0.0200	inv	0.0050
eq	0.008	0.0086	0.0066	0.0105	inv	0.0050

警告: エクスポートされたイメージには座標軸ツール バーが表示されています。イメージから座標軸ツール バーを削除するには、再度エクスポートしてください。

Total computing time : 0h45m38s

Note: warning(s) encountered in MATLAB/Octave code

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